Case F

Mr. Robert Miller, 64M — Day 2 Post-CABG (Median Sternotomy) | 30 pack-year smoking · Mild baseline COPD · Sternal pain / "clicking" · Chest drain + telemetry · Shallow apical breathing · Weak guarded cough · Goal: prevent PPCs & early mobilisation

Cardiorespiratory Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Post-sternotomy pain → splinting → basal atelectasis
- Reduced chest wall compliance post-surgery
- Unable to clear secretions effectively
- Bed-bound → DVT risk, deconditioning
- Mobility limited by chest drain and telemetry attachments

■■ RED FLAGS

- only with coughing may be gas or tissue movement and is more likely
- benign early post-operatively.
- "Can you take a deep breath, or does pain stop you before you get there?"
- I ask this to establish current
- ventilatory capacity and identify pain-splinting behaviour. Inadequate
- tidal volume drives atelectasis — the primary mechanism of
- post-operative pulmonary complications.
- "Before the operation, did you have a regular morning cough, bring up sputum, or get breathless with everyday activities?"

■ PATHOPHYSIOLOGY (KEY POINTS)

- Surgical mechanism of PPC: Median sternotomy divides the sternum and retracts the chest wall, causing direct thoracic cage trauma.
- Atelectasis cascade: Surfactant dysfunction from bypass, diaphragmatic splinting from pain, and impaired cough reflex allow distal airways to collapse.
- Gravitational forces compound basal zone collapse in supine positioning.
- The result is ventilation/perfusion (V/Q) mismatch, hypoxaemia, and a fertile environment for post-operative pneumonia.
- COPD additive risk: Robert's mild COPD means he has reduced mucociliary clearance, increased mucus production, and already-compromised small airway function at baseline.
- General anaesthesia and surgical stress further impair these defences, placing him at significantly greater risk of PPCs than a non-COPD patient.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observations

RR: expected 18–24/min (splinting) SpO■: target ≥94% on O■ HR: telemetry — watch for arrhythmia Sternal wound: dressing intact, no separation Chest drain: character and volume of output

Auscultation Findings Expected

Diminished or absent breath sounds bilaterally at bases — basal atelectasis Fine inspiratory crackles — alveoli opening on deep inspiration (positive sign of response) Coarse crackles if secretion retention present — more likely in COPD patient Upper zone breath sounds preserved — consistent with apical breathing pattern

Chest Expansion Measurement

What it measures repeatable measure of breathing restriction from sternal pain and splinting, and quantifies improvement after breathing exercises — guiding which lung zones to target."

Incentive Spirometry Assessment

What it measures Sustained maximal inspiratory effort; degree of atelectasis and inspiratory muscle weakness; visual biofeedback tool How to perform Patient in high Fowler's (60–80°). Apply sternal support pillow first. Instruct

■ TREATMENT PLAN

- prerequisite for effective post-sternotomy physiotherapy. Uncontrolled
- pain directly causes shallow breathing, guarded cough, and increased PPC
- risk. I need to time therapy 30–45 min after analgesia peak.
- "When you say the sternum is 'clicking' — can you tell
- me more? Does it click with every breath, only with movement, or only
- when you cough?"
- I ask this to assess sternal stability.
- Clicking with every breath or at rest suggests possible dehiscence — a
- seated or semi-reclined. Tape measure at 4th intercostal space

• (mid-chest) and at xiphisternum (lower chest). Record circumference at
• full expiration and full inspiration; calculate difference. Normal ≥5
• cm. Assess symmetry with bilateral hand placement — unequal expansion